Model Selection & Iteration

GeoPython — Quick Reference Handout 05

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1. The Model as a Hypothesis

Choosing a model is not a final decision — it is a hypothesis. Like any hypothesis, it will be tested against evidence (your test data), refined based on failure patterns, and replaced when a better explanation emerges.

> **!** *There is no "perfect" model out of the box. The first model you choose will almost certainly fail in a specific, informative way. That failure is data, not defeat.*

2. Common Models and Their Failure Modes

Model	Strength	Typical Failure Mode in Geoscience
Linear Regression	Fast, interpretable	Misses non-linear patterns; poor on seasonal "cliffs"
Random Forest	Handles non-linearity, robust	Cannot extrapolate beyond training range; slower
KNN	Simple, no assumptions	Poor performance in high dimensions; needs scaling
SVR	Effective in high dimensions	Kernel and parameter tuning required; computationally heavy
Gradient Boosting	High accuracy	Easy to overfit; longer training time; many hyperparameters

3. The Iteration Workflow

Model iteration follows a consistent loop. Stop when your test metrics and residuals indicate valid predictions — not when you've tried every model.

Step	Action	Output
1. Baseline	Fit simplest model (Linear Regression)	Benchmark score + residual pattern
2. Diagnose	Plot residuals; check for systematic error	Identify failure type (non-linear? seasonal?)
3. Feature engineering	Add/transform features to address the failure	Revised feature set
4. Try next model	Fit a more complex model on revised features	New score — compare to baseline
5. Validate	Check test metrics AND Durbin-Watson	Accept or continue iterating

> *If a more complex model gives only marginal improvement, prefer the simpler one. Complexity that does not improve predictions is overfitting in disguise.*